

A Packing of 19 Four-Spaces in $\mathbb{R}^{12}$

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October 9, 2026

Abstract

We give 19 4-dimensional subspaces of $\mathbb{R}^{12}$ whose smallest squared chordal distance is at least 2.814438. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.806619, obtained by deleting 1 of the 20 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 19 subspaces of dimension 4 in $\mathbb{R}^{12}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.814438 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 12$, $n = 4$. The cell $N = 19$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.806619, obtained by deleting 1 of the 20 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(12 - 4)/12 \cdot 19/18 = 2.814815$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m12n4N19.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
255858048837541085926590724929643359949704470322155623655086058726981541
385623708589276465371142759643795177170497186270020057665785449505642949
737758760952129210037393670868351679361157581925
```

denominator:

```
909090909089277261696823675762475886274997182657357493686399135088948360
387398023737538075131610140507357240447666789654996321714687225756356504
63430660481298070835727062032095303768243745646
```

B Coordinates

19 blocks of 12 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m12n4N19.txt`.

```
434665531707 274336746924 607335229963 57389799883 19926648576 103217372969 -245017331755 -50156927876241560186859 -364855897085 -355045027580 76722134937
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